## LAPORAN ANALISIS PRAKTIKUM PEMROGRAMAN BASIS DATA

## KUIS PERTEMUAN 4 – STRUKTUR KONTROL PERCABANGAN

**Nama:** Nur Fahila Dwi Irfani.D
**NIM:** IK2411031
**Prodi:** informatika
**Mata Kuliah:** Pemrograman Basis Data

# 1.Deskripsi Kasus

Pada tugas praktikum ini dibuat beberapa *stored procedure* menggunakan struktur kontrol percabangan **IF–THEN–ELSE** pada MySQL untuk menyelesaikan tiga kasus utama. Kasus pertama yaitu pengecekan status stok barang berdasarkan jumlah stok yang tersedia, sehingga sistem dapat menentukan apakah stok habis, hampir habis, tersedia, atau aman. Kasus kedua yaitu perhitungan diskon belanja berdasarkan total transaksi, di mana besarnya diskon ditentukan oleh nominal pembelian pelanggan. Kasus ketiga yaitu penentuan predikat mahasiswa berdasarkan nilai yang diperoleh serta status kelulusannya. Ketiga kasus tersebut bertujuan menunjukkan bagaimana percabangan dapat digunakan untuk pengambilan keputusan otomatis dalam basis data.

**1. Pengecekan Status Stok**

Sistem dibuat untuk mengecek kondisi stok barang dan menentukan status:

- Habis
- Hampir Habis
- Tersedia
- Stok Aman

Data produk yang digunakan:

| Produk | Stok |
|---|---|
| Pensil | 8 |
| Buku | 15 |
| Penghapus | 2 |
| Pulpen | 30 |
| Spidol | 12 |

**2. Perhitungan Diskon**

Procedure dibuat untuk menghitung diskon pelanggan berdasarkan total pembelian:

| Total Belanja | Diskon |
|---|---|
| ≥ 1.500.000 | 20% |
| ≥ 700.000 | 12% |
| ≥ 300.000 | 7% |
| lainnya | 1% |

**3. Predikat Mahasiswa**

Procedure digunakan untuk menentukan predikat nilai mahasiswa serta status kelulusan.

| Nilai | Predikat |
|---|---|
| 95+ | Sangat Memuaskan |
| 85–94 | Memuaskan |
| 75–84 | Baik |
| 65–74 | Cukup |
| <65 | Kurang |

Status:

- ≥75 Lulus
- <75 Tidak Lulus

# 2.PENJELASAN LOGIKA PERCABANGAN

**1 IF ELSE Status Stok**

Logika percabangan:

```
IF p_stok =2 THEN
'Habis'

ELSEIF p_stok BETWEEN 3 AND 7 THEN
'Hampir Habis'

ELSEIF p_stok BETWEEN 8 AND 25 THEN
'Tersedia'
```

ELSE

'Stok Aman'

Percabangan ini digunakan untuk mengelompokkan kondisi stok berdasarkan rentang nilai.

**2 IF ELSE Diskon**

Logika diskon:

IF p_total >=1500000 THEN
20%

ELSEIF p_total >=700000 THEN
12%

ELSEIF p_total >=300000 THEN
7%

ELSE
1%

Logika digunakan untuk menentukan potongan harga secara otomatis.

**3 IF ELSE Predikat Mahasiswa**

IF p_nilai >=95 THEN
Sangat Memuaskan

ELSEIF p_nilai >=85 THEN
Memuaskan

ELSEIF p_nilai >=75 THEN
Baik

ELSEIF p_nilai >=65 THEN
Cukup

ELSE
Kurang

Kelulusan:

IF p_nilai >=75 THEN
Lulus
ELSE
Tidak Lulus

# 3.LANGKAH PENGERJAAN

1. Membuat database dan tabel produk.
2. Menginput data contoh.
3. Membuat stored procedure:

- cek_status_stok
- hitung_diskon
- cek_predikat_mahasiswa

4. Melakukan pengujian menggunakan:

CALL cek_status_stok(4);

CALL hitung_diskon(800000);

CALL cek_predikat_mahasiswa(88);

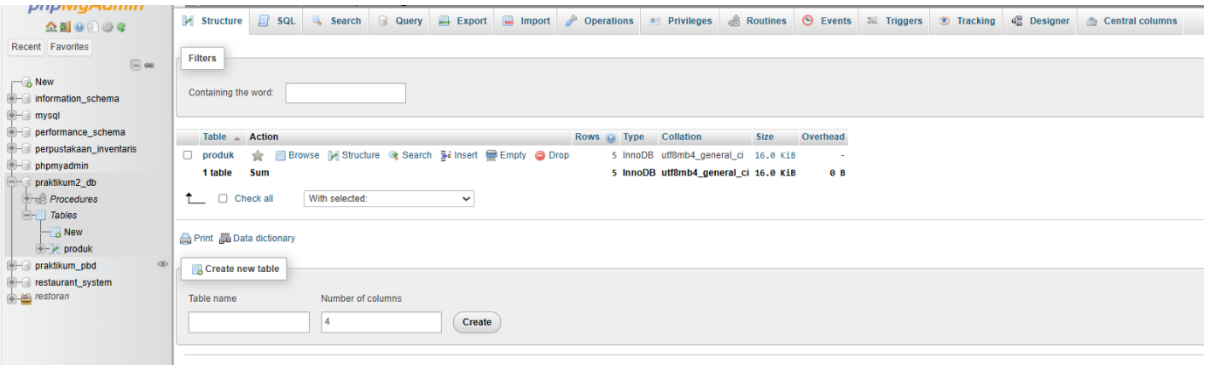
5. Menganalisis output yang dihasilkan.

# 4.SCREENSHOT HASIL EKSEKUSI

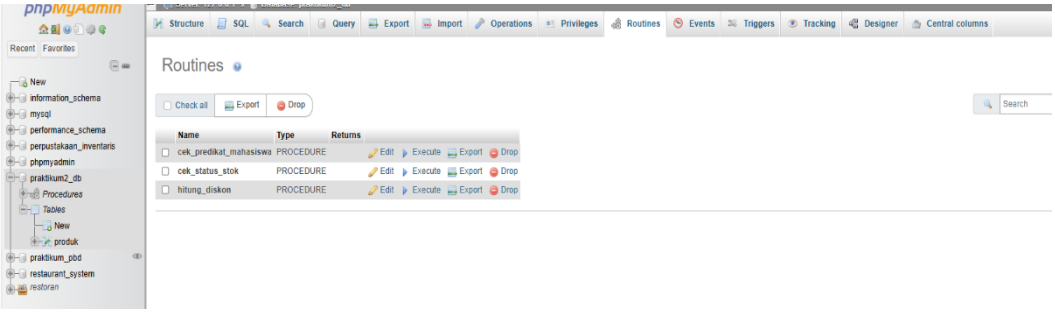
**a. Screenshot Kode Dijalankan**

Tempel screenshot:

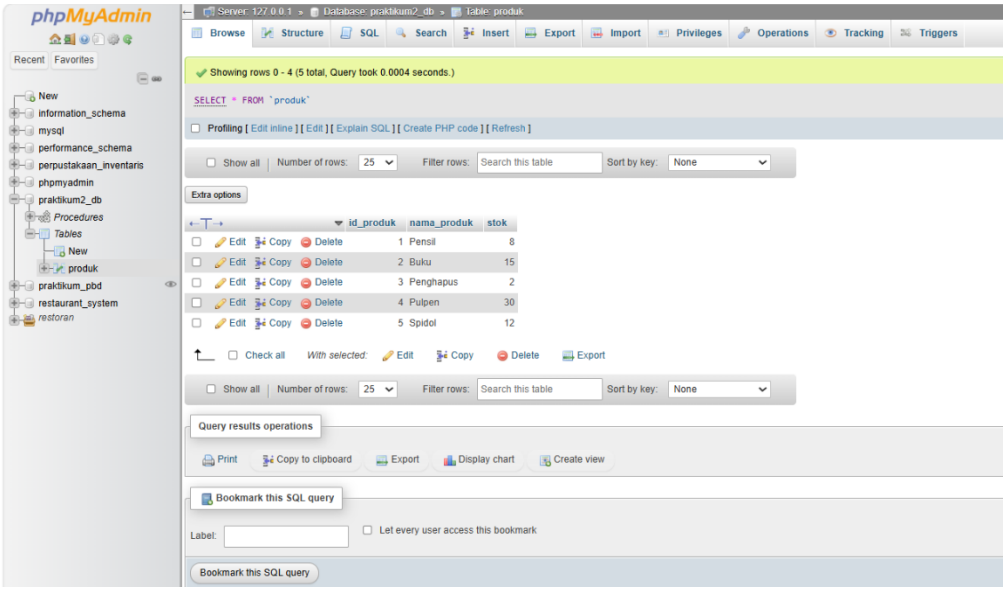
- Pembuatan tabel




- Pembuatan procedure




**b. Hasil Data Produk Berikut merupakan hasil data yang telah dimasukkan.**




*cek_status_stok*

*CALL cek_status_stok(8);*

*CALL cek_status_stok(15);*

*CALL cek_status_stok(2);*

*CALL cek_status_stok(30);*

*CALL cek_status_stok(12);*

Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)

```
CALL cek_status_stok(8);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| stok | status |
|---|---|
| 8 | Tersedia |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Showing rows 0 - 0 (1 total, Query took 0.0006 seconds.)

```
CALL cek_status_stok(15);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| stok | status |
|---|---|
| 15 | Tersedia |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
CALL cek_status_stok(2);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| stok | status |
|---|---|
| 2 | Habis |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL cek_status_stok(30);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| stok | status |
|---|---|
| 30 | Stok Aman |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

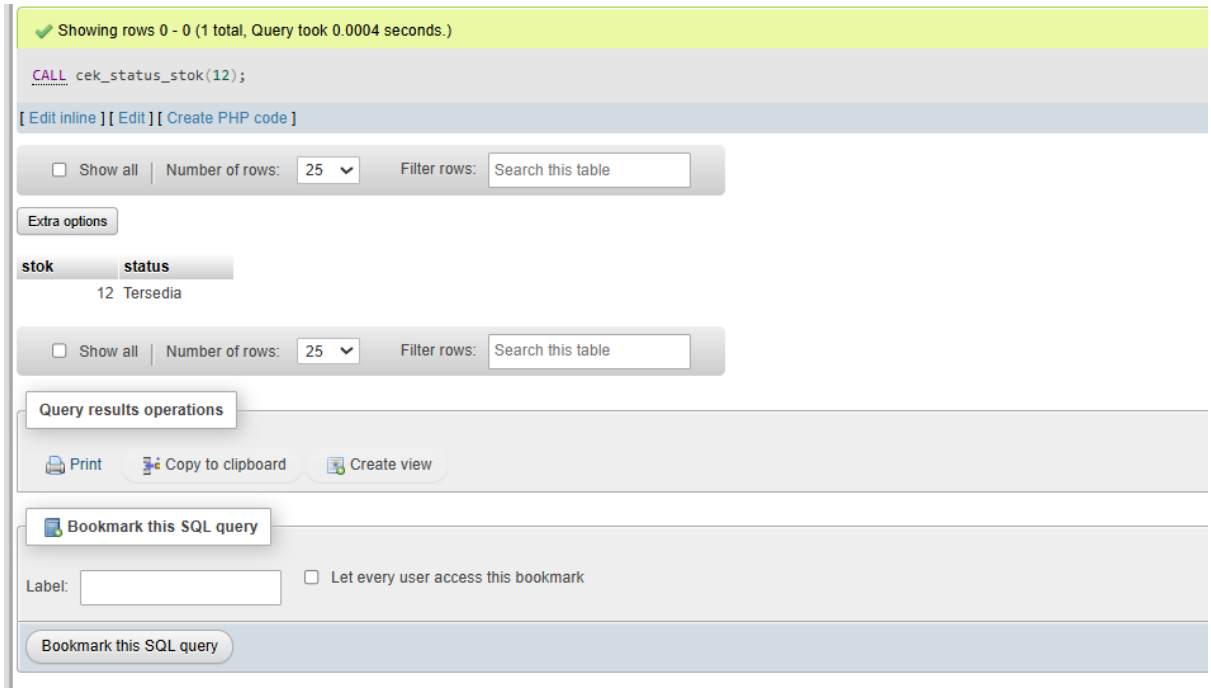
Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.




**f. hitung_diskon**

**Perhitungan Diskon**

**Aturan Diskon**

**Total Belanja**

**≥1.500.000**

**Diskon 20%**

**≥700.000**

**Diskon 12%**

**≥300.000**

**Diskon 7%**

**Lainnya 1 %**

## Total Belanja

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds.)

```
-- Latihan 3 CALL hitung_diskon(1500000);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| total_belanja | diskon_persen | jumlah_diskon | total_bayar |
|---|---|---|---|
| 1500000.000 | 20 | 300000.000 | 1200000.000 |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL hitung_diskon(700000);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| total_belanja | diskon_persen | jumlah_diskon | total_bayar |
|---|---|---|---|
| 700000.000 | 12 | 84000.000 | 616000.000 |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

**Total Belanja**

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL hitung_diskon(300000);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| total_belanja | diskon_persen | jumlah_diskon | total_bayar |
|---|---|---|---|
| 300000.000 | 7 | 21000.000 | 279000.000 |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

**Total Belanja**

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL hitung_diskon(100000);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| total_belanja | diskon_persen | jumlah_diskon | total_bayar |
|---|---|---|---|
| 100000.000 | 1 | 1000.000 | 99000.000 |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

**Predikat Mahasiswa**

nilai : 88
predikat : Memuaskan
status : Lulus

Showing rows 0 - 0 (1 total, Query took 0.0009 seconds.)

```
CALL cek_predikat_mahasiswa(95);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| nilai | predikat | status |
|---|---|---|
| 95 | Sangat Memuaskan | Lulus |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL cek_predikat_mahasiswa(85);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| nilai | predikat | status |
|---|---|---|
| 85 | Memuaskan | Lulus |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
CALL cek_predikat_mahasiswa(75);
```

[ Edit inline ] [ Edit ] [ Create PHP code ]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

| nilai | predikat | status |
|---|---|---|
| 75 | Baik | Lulus |

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

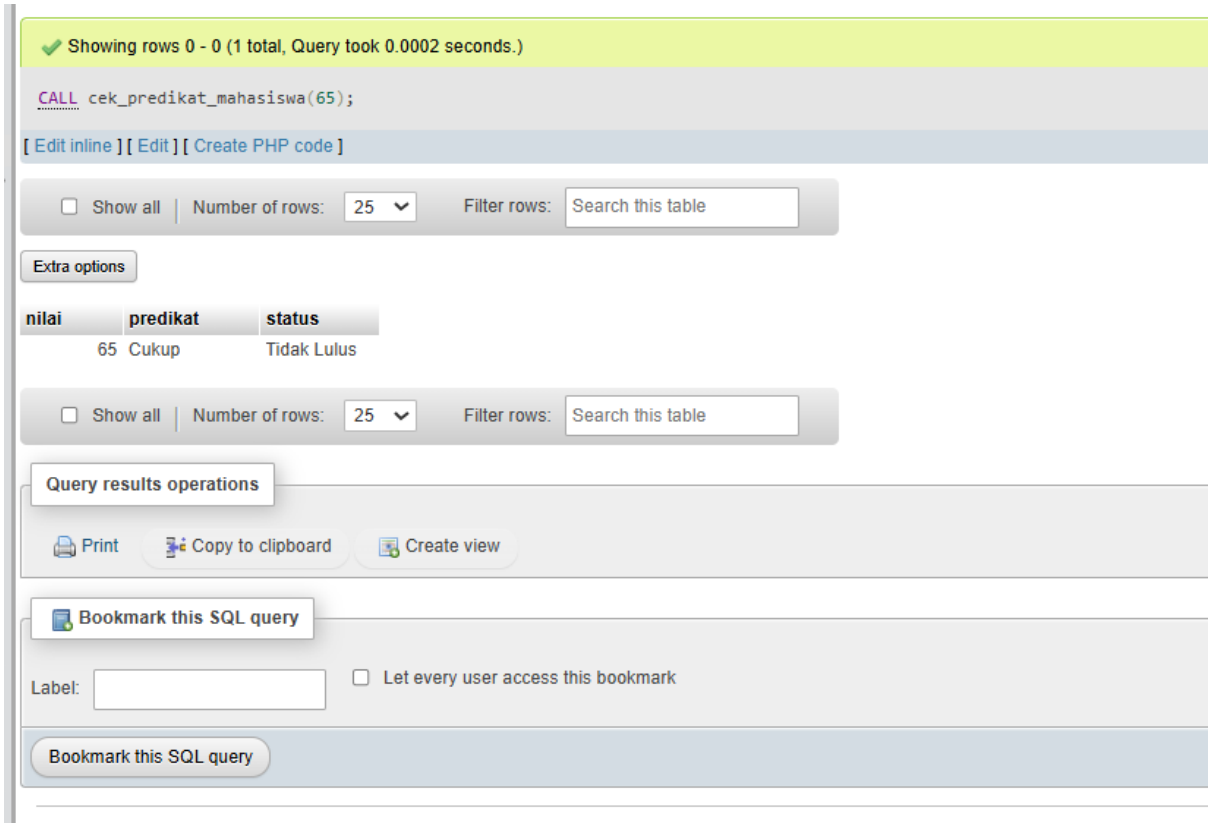
Print Copy to clipboard Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.




# 5.ANALISIS OUTPUT

**Analisis Kasus Stok**

Input 4 menghasilkan status *Hampir Habis* karena berada pada rentang 3–7.

**Analisis Kasus Diskon**

Input 800000 memperoleh diskon 12% sesuai aturan percabangan.

**Analisis Kasus Predikat**

Nilai 88 menghasilkan predikat *Memuaskan* dan status *Lulus* karena memenuhi syarat kelulusan.

Semua output menunjukkan logika percabangan berjalan sesuai ketentuan.

# 6.KESIMPULAN

Berdasarkan praktikum yang telah dilakukan, struktur kontrol percabangan IF-THEN-ELSE pada MySQL berhasil diterapkan dalam stored procedure untuk pengambilan keputusan otomatis. Implementasi pada pengecekan stok, perhitungan diskon, dan penentuan predikat mahasiswa menghasilkan output yang sesuai dengan logika yang dirancang. Dengan demikian tujuan praktikum tercapai dan penggunaan percabangan dalam pemrograman basis data dapat dipahami dengan baik.